NIKHIL MAHADEVAN

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Education

University of Georgia | Athens, GA **Aug 2024 - May 2028**

B.S. in Electrical Engineering, B.S. in Computer Engineering **GPA: 3.93/4.0**

Relevant Coursework: Logic Design, Electronics, Sensors and Transducers, Informatics, Linear Systems, Embedded Systems

Experience

Flight Software Engineer | University of Georgia Small Satellite Research Laboratory **Sep 2026 - Present**

- Developing flight software in C under Zephyr RTOS on an STM32 MCU for CubeSat communications and command handling
- Implementing serial drivers and communication protocols (UART, RS-485) between onboard satellite subsystems
- Validating multi-subsystem firmware and bus communication protocols using Renode hardware emulation before bench testing

Embedded Systems Research Intern | University of Georgia Vertically Integrated Projects for Research **Aug 2026 - Present**

- Writing C++ drivers (I2C, SPI) for IMU, temperature, and humidity sensors on an ATmega328P-based wearable heat-stress tracker for canines
- Implementing Bluetooth communication to stream sensor data to a host application in real time

Electronics Lab Assistant | University of Georgia School of Engineering & Computer Engineering **Aug 2026 - Present**

- Instructing 100+ students across two labs on analog filters, oscillators, transformers and BJT amplifiers with SPICE verification
- Guiding students with TTL circuit builds and benchtop debugging using oscilloscopes, logic analyzers and multimeters

Engineering Peer Tutor | University of Georgia **Sep 2025 - May 2026**

- Tutored 35+ students across courses mainly focused on Logic Design (CSEE 2220) and Circuit Analysis (ECSE 2170)

Projects

Edge AI Wake-Word Engine | github.com/nmahadevan12/Edge-AI-STM32 **Aug 2026**

- Engineered register-level C firmware on an STM32L4 (ARM Cortex-M4) for real-time PDM audio acquisition
- Implemented a DC-blocking filter and energy detector to isolate speech buffers and reject background noise
- Developed a 921.6 kbps UART driver with SysTick interrupt-driven timestamping to stream audio to a host PC
- Trained an MFCC-based wake-word model in Python (92% accuracy) to detect “Hey STM32”

2-Axis Light Tracker (ELEE 4230) | Project video | github.com/nmahadevan12/Light-Tracker **Jun 2026 - Jul 2026**

- Built a battery-powered dual-axis light tracker in bare-metal C on an ATmega328P using MOSFET H-bridges to drive motors
- Integrated an LDR array with Hall-effect and potentiometer feedback for tilt limit protection
- Implemented a fixed-point Kalman filter to smooth ADC readings, reducing peak noise from 3 LSBs to < 1 LSB
- Engineered a register-level UART driver (9.216 MHz crystal) to stream real-time system telemetry
- Designed a four-layer PCB in KiCAD for the ATmega328P microcontroller

Multifunction Instrumentation System (ECSE 2920) | github.com/nmahadevan12/MIS **Jan 2026 - Apr 2026**

- Built a Raspberry Pi DMM and function generator in Python powered by a ±12V split-rail supply with active virtual ground
- Designed a discrete SAR ADC via SPI digipot and comparator for voltmeter, ohmmeter, and DC reference
- Improved DC resolution 6x and voltmeter precision 2x via digipot calibration
- Designed a 4-layer mixed-signal Pi HAT PCB with dedicated power and ground planes

Technical Skills

Languages: C, C++, ARM Assembly, Python, Java, MATLAB

Architectures & Hardware: ARM Cortex (STM32), AVR (ATmega), ESP32, Raspberry Pi, SPI, I2C, UART, PWM

Tools & Equipment: Altium Designer, Autodesk Fusion, KiCad, SPICE, Git, Oscilloscopes, Logic Analyzers, Multimeters, Soldering

Honors/Memberships

UGA IEEE MicroMouse, UGA Dean’s List, Zell Miller Scholar, UGA Outdoor Club, Eagle Scout (2022)